Volume III · Chapter 17 — Capstone Project · Labs

> Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-III-Chapter-17-context.md` for the AI interaction log.

Prerequisites: all of Volumes I–III.

Learning outcomes — the student can: independently design, build, evaluate, and deploy an end-to-end medical-AI system for a real clinical problem; document decisions and safety; present with evidence to clinical stakeholders.

Project (hands-on) — 60 h:

#	Milestone	h
17a	Problem definition & data/knowledge sourcing	8
17b	Build knowledge base / RAG pipeline	12
17c	Train/adapt the model (QLoRA/DPO as needed)	14
17d	Evaluation & safety/fairness assessment	10
17e	Deploy locally with FHIR-style I/O (inference server)	10
17f	Documentation, model card & final	6

#	Milestone	h
	presentation	

Example project: "Clinical Trial Matching Assistant" (plan §5.2). **Datasets/tools:** the full stack from Volumes I–III. **Assessment:** end-to-end system (**50%**); evaluation & safety documentation (**25%**); presentation/defense (**25%**). **Key decisions:** the full stack of trade-offs — RAG vs. fine-tune, hardware, scope vs. safety. **References:** plan §5.2; §13 (all). **Hours:** Theory **6** + Project **60** = **66**.